Advanced AI Campus Hackathon

Smart Campus, Better Society

Expanded Problem Statement Catalogue — Revised

Difficulty scale: 1/10 = introductory, 10/10 = extremely challenging.

Evaluation suggestion: Social impact 25% | Technical implementation 25% | Innovation 20% | UI/UX 15% | Practical feasibility 15%

Original Problem Statements

PS-01 Student Dropout Early Warning System

Use simulated attendance, academic performance, and engagement data to identify students who may need additional support. Create explainable, privacypreserving predictions and equitable intervention recommendations.

PS-02 Intelligent Campus Disaster Response

Build a platform to simulate fires, floods, earthquakes, and other campus emergencies. Implement dynamic evacuation routing that adapts to hazards, crowd density, and accessibility needs.

PS-03 Campus Food Demand Prediction

Forecast daily mess and canteen demand using historical meals, holidays, exam schedules, and attendance estimates. Reduce food waste while avoiding shortages under uncertain demand.

PS-04 AI Campus Accessibility Auditor

Analyze campus maps and infrastructure images to identify obstacles faced by people with disabilities. Recommend accessible routes and prioritize infrastructure improvements.

PS-05 Campus Cybersecurity Threat Detection

Create a dashboard that monitors a simulated campus network for suspicious activity. Detect anomalous logins, phishing indicators, and attack patterns with low false-positive rates.

PS-06 Campus Resource Allocation Optimizer

Allocate classrooms, laboratories, equipment, and study areas based on competing requirements. Solve scheduling conflicts under capacity, fairness, accessibility, and availability constraints.

PS-07 Privacy-Preserving Harassment Pattern Detection

Design a confidential reporting system that helps authorized staff identify potentially related incidents. Find patterns without exposing complainants, making automated accusations, or bypassing human review.

PS-08 Campus Energy Optimization

Predict electricity demand in hostels, labs, and academic buildings. Use occupancy, weather, and simulated meter data to reduce consumption while maintaining essential services.

PS-09 Campus Misinformation Detection

Help students check potentially false college announcements against verified institutional information. Provide evidence-linked assessments, uncertainty labels, and source traceability.

PS-10 Emergency Medical Coordination

Coordinate verified emergency requests with campus responders and healthcare contacts. Optimize assignment and routing using incident severity, availability, and travel time.

PS-11 Campus Sustainability Decision Engine

Evaluate the environmental impact of campus operations across energy, water, waste, and transport. Recommend a cost-constrained portfolio of improvements using multi-objective optimization.

Additional High-Impact Problem Statements

PS-12 AI-Based Mental Health Support and Crisis Detection

Develop a privacy-preserving platform that identifies students at risk of severe stress, burnout, or self-harm using voluntary surveys, academic patterns, and help-seeking behavior. Detect risk without diagnosis, protect privacy, minimize false alarms, and connect students to verified support.

PS-13 Women's Safety and Safe Route Optimizer

Recommend safer walking routes using lighting, crowd density, emergency phones, security coverage, and historical incident data. Optimize routes in real time while protecting sensitive location data and avoiding victim-blaming.

PS-14 AI Sign-Language and Speech Accessibility Assistant

Create a real-time tool that converts sign language to text or speech and spoken language into visual instructions. Support diverse signing styles, regional languages, limited training data, and low-latency operation.

PS-15 Assistive Navigation for Visually Impaired Students

Develop an AI navigation assistant using computer vision, maps, voice instructions, and obstacle detection. Provide safe, accurate, real-time guidance and identify temporary obstacles.

PS-16 Scholarship and Financial-Aid Eligibility Engine

Identify students who may qualify for scholarships, fee support, food assistance, or emergency funds. Match complex eligibility rules while protecting financial data and preventing discrimination.

PS-17 Campus Food Rescue and Redistribution Platform

Predict surplus food from hostels and canteens and coordinate safe redistribution to students, shelters, or community organizations. Forecast surplus, optimize collection routes, and monitor food safety.

PS-18 Student Accessibility Accommodation System

Convert confidential disability-related accommodation requests into coordinated academic, transportation, examination, and housing support. Prioritize urgent needs, detect service gaps, and provide explainable recommendations.

PS-19 Human Trafficking and Exploitation Risk Awareness System

Detect suspicious patterns in simulated campus or community reports without automatically accusing individuals. Analyze weak signals, preserve anonymity, reduce bias, and require human verification before escalation.

PS-20 AI-Based Academic Integrity and Fairness Monitor

Detect suspicious academic behavior while protecting students from unfair automated punishment. Distinguish collaboration from cheating, detect bias, provide evidence, preserve privacy, and keep humans in the decision loop.

PS-21 Community Blood and Organ Donation Coordinator

Predict blood demand, verify donors, match compatible donations, and coordinate urgent requests. Handle real-time matching, medical privacy, availability uncertainty, and emergency prioritization.

PS-22 Campus-to-Employment Opportunity Matcher

Connect students—especially first-generation, rural, or economically disadvantaged learners—with suitable internships, jobs, and skill-development programs. Match skills to opportunities, identify skill gaps, explain recommendations, and prevent bias.

Recommended Highest-Impact Choices

For maximum technical challenge and social value, consider PS-12, PS-14, PS-15, PS-19, PS-21 and PS-22 These combine AI complexity, measurable social benefit, safety considerations, privacy requirements, and strong opportunities for innovation.

General Organizer Notes

• Provide common or simulated datasets and surprise test cases.

• Judge working functionality, safety, fairness, and measurable outcomes—not chatbot integration alone.

• Require human review for high-risk decisions involving health, safety, harassment, finances, or academic consequences.

• Teams should document limitations, false positives, privacy protections, and responsible-AI safeguards.